Imperial College London

Robot Intelligence Lab

Final Report for MSc Project

New gait for a knee-less bipedal robot: SLIDER's RotoGait

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1. Abstract

This research presents a new gait for the knee-less bipedal robot SLIDER where the legs fully rotate around the body: RotoGait. The trajectory followed by SLIDER during the movement as well as a simplified implementation in simulation are presented. The work also shows that because of shortcomings in the current whole body controller used, SLIDER cannot yet use its own control for this new movement. The dynamics of RotoGait will have to be used in the future to make the controller compliant with the movement. Furthermore, the present work shows that for the tested speeds, RotoGait is not more efficient than normal gait but shows every indication that with an optimal trajectory and footstep position RotoGait could be more efficient than normal gait at higher speeds. Optimization of swingfoot trajectory and modification of the footstep planner to optimize footstep position based on RotoGait movement should therefore be added in the future.

2. Acknowledgements

The author would like to thank Dr. Petar Kormushev for welcoming her into the Robot Intelligence Lab and the great opportunity to work on a new gait for Imperial College London's first bipedal robot.

The author is also truly grateful to Digby Chappell and Ke Wang for all their help, support and guidance throughout the project. Their encouragements and expertise have been invaluable to realising the present work.

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3. Introduction

Imperial College London's first bipedal robot has been under development at the Robot Intelligence Lab: SLIDER, shown in *Figure 1*. Most bipedal robots are designed to resemble humans, with actuated hips, knees and ankles as Boston Dynamics' ATLAS® [1] and Agility Robotics' CASSIE and DIGIT [2]. Because of this design, the weight of the knee actuator that has to be mounted on the robot's leg leads to making a choice between using powerful motors to move the robot or sacrifice its locomotion ability [1], [3]. To compensate for this and have more lightweight legs, some robots where simplified to consider straight legs to avoid the additional weight of the knees [4], [5].

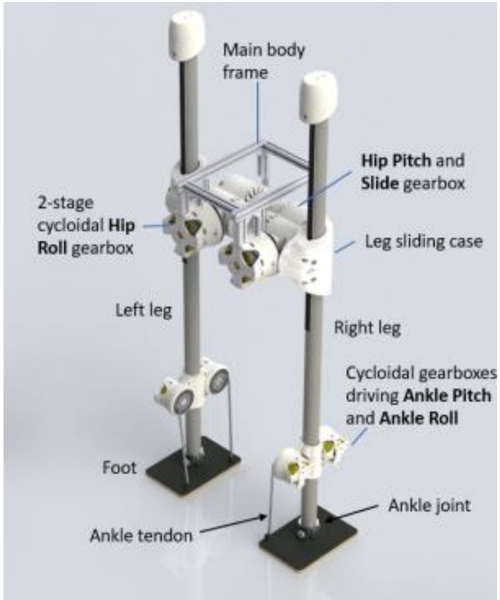

Figure 1 - Model of the SLIDER robot [7]

Nevertheless, this simplification often removes one or more joints without compensating for the lost degree of freedom (DoF) of those joints, effectively limiting the motion of the robot. SLIDER resolves those issues, by removing the knee joint but adding a vertical sliding motion of the hip to keep the knee DoF, as shown in the model of SLIDER in *Figure 1*. This design allows to remove the weight of the knee actuation to have a lightweight leg, albeit with added complexity at the hip, without sacrificing the robots DoF [6], [7]. With the hip roll, pitch and sliding and with the ankle roll and pitch SLIDER has proven to be able to walk both in simulation and in the physical world [7], [8], confirming the potential capacities of this new design. Not only does it allow to produce a robot with lightweight legs and that presents an agility that doesn't require much power, but it also allows us to consider a whole new gait, called RotoGait.

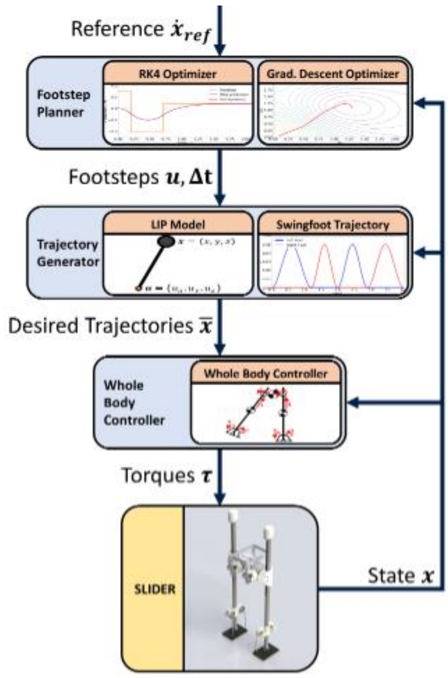

Figure 2 - SLIDER's control diagram [9]

In normal walking, one leg is in contact with the ground (called stance or support leg and will from now on be called support leg) and only rotates around that contact point while the other leg (called swing leg) swings forward until it hits the ground and becomes the new support leg, as shown in *Figure 3*. But as SLIDER doesn't have knees, it can perform a full rotation of the hip pitch joint and use both end of each leg as feet instead of swinging the swing leg forward as shown in *Figure 4*. A footstep planner is used to compute the footsteps' location in real time [7] and a whole body controller is used to control SLIDER's movements [9], as shown in *Figure 2*. The whole body controller works by minimizing the error between desired and actual acceleration of the Center of Mass (CoM) and the swingfoot. This objective function is constrained by the rigid body dynamics, a friction model for non-slippery contact, torque limits and a force selection matrix for the force applied to the feet (see Chappell and Wang (2021) [9] for more details). The only difference between the control of the two gaits would therefore be the trajectory followed by the swing leg.

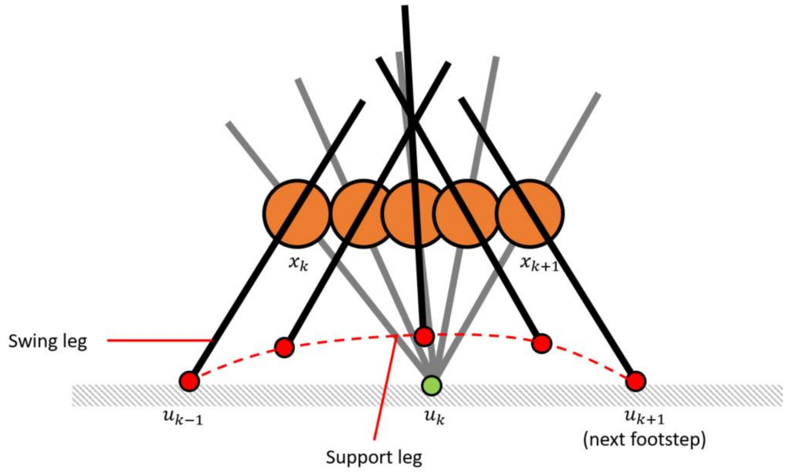

Figure 3 - Diagram of normal gait trajectory [10]

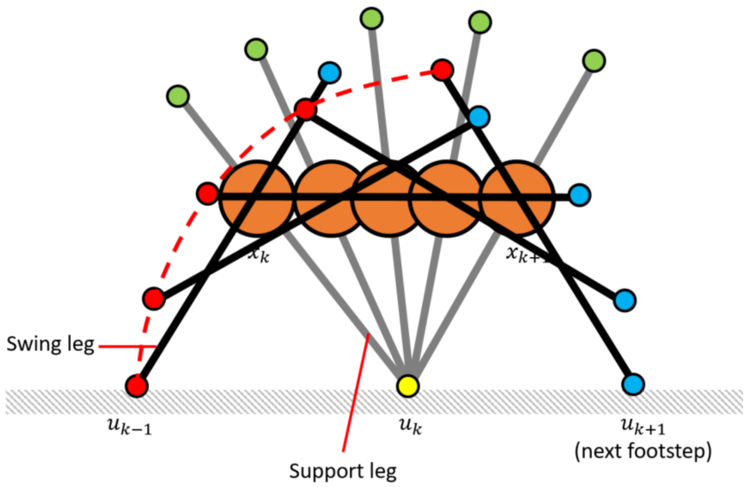

Figure 4 - Diagram of RotoGait's trajectory [10] *(Configuration one)*

As this movement is a new concept there is no published research on it, to the knowledge of the author, this concept yet appears in the TARS robot in Interstellar [11]. Nevertheless, a similar movement is the one of the rimless wheel [12], [13]. The rimless wheel consists of several spokes evenly spaced out to form wheels on both side of a robot, as shown in *Figure 5*. This concept is widely spread and has proven to allow stable gait [14]. However, rimless wheeled robots present some key differences with SLIDER's RotoGait. Firstly, most rimless wheel robots have at least 6 spokes on either side of the body with some robots only having 3 on each side [13], while SLIDER only has the 2 ends of both leg constituting only 2 spokes on each sides, which has never been seen in a robot to the knowledge of the author. Secondly, rimless wheels operate with a fixed angle between both wheels with usually one wheel position complementing the spokes' position of the other wheel or both wheels having same spokes orientation. SLIDER on the other hand operates both legs separately and is not bound to a particular angle between the two legs. Thirdly, most rimless wheel don't have a possibility for a sliding motion of the spokes with respect to the body of the robot, even if some have passive sliding motion of the spokes during movement [12], [15]. At the exception of Impass [16] which has an actuated sliding motion of the spokes but requires an additional contact point with the ground, SLIDER as this unique ability to slide the legs at the hip joint.

Figure 5 - Representation of a rimless wheel [12]

As one hypothesis of this work is that RotoGait could be more efficient than normal gait for certain velocities, the power consumption of the movement needs to be studied. One way to determine the power consumption is to consider it equal to the motor's power consumption [17], but this ignores the electrical energy to maintain a specific joint torque and cannot be measured in simulation. Many studies integrate the electrical power consumption, either entirely [18], [19], by taking its absolute value (or the square of the absolute value) [20], [21], or even just considering the positive values [22]. This would give a good estimate of the total power consumption of SLIDER, but cannot be applied to the simulation environment. Similarly, to the motor's power consumption, the mechanical power consumption can be determined by integrating the product of the joint's torque and velocity [23], [24]. This method can be applied in the simulation environment but fails again to consider the electrical energy to maintain a torque.

This RotoGait could allow SLIDER to achieve higher velocities than normal walking and may be more efficient than normal walking at higher speeds. The main objective of this project is therefore to see if RotoGait can be implemented on SLIDER in simulation. The second objective is to compare the power consumption of RotoGait against normal gait to determine if and when it is more efficient.

4. Methods

As explained in the introduction, the only difference in the control of SLIDER for RotoGait compared to the diagram of *Figure 2*, is the swingfoot trajectory generated. It is modified to get the foot to the new position planned by the footstep planner following the trajectory in *Figure 4* instead of the one in *Figure 3*.

4.1. Trajectory generator

The swingfoot's position follows a fifth order polynomial as defined in the following equations:

$$x(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 + a_5 t^5 \tag{1}$$

With

$$a_0 = x_i \tag{2}$$

$$a_1 = \dot{x}_i \tag{3}$$

$$a_2 = \frac{1}{2}\ddot{x}_i \tag{4}$$

$$a_3 = \frac{1}{2T^3}\left[20\left(x_f - x_i\right) - \left(8\dot{x}_f + 12\dot{x}_i\right)T - \left(3\ddot{x}_i + \ddot{x}_f\right)T^2\right] \tag{5}$$

$$a_4 = \frac{1}{2T^4}\left[30\left(x_i - x_f\right) + \left(14\dot{x}_f + 16\dot{x}_i\right)T + \left(3\ddot{x}_i - 2\ddot{x}_f\right)T^2\right] \tag{6}$$

$$a_5 = \frac{1}{2T^5}\left[12\left(x_f - x_i\right) - 6\left(\dot{x}_f + \dot{x}_i\right)T - \left(\ddot{x}_i - \ddot{x}_f\right)T^2\right] \tag{7}$$

Where $x_i, \dot{x}_i, \ddot{x}_i$ are the starting position, velocity and acceleration, $x_f, \dot{x}_f, \ddot{x}_f$ are the ending position, velocity and acceleration of the movement, and T is the duration of the movement. This also leads to the following velocity and acceleration equations:

$$\dot{x}(t) = a_1 + 2 * a_2 t + 3 * a_3 t^2 + 4 * a_4 t^3 + 5 * a_5 t^4 \tag{8}$$

$$\ddot{x}(t) = 2 * a_2 + 6 * a_3 t + 12 * a_4 t^2 + 20 * a_5 t^3 \tag{9}$$

As a result, $x_i, \dot{x}_i, \ddot{x}_i, x_f, \dot{x}_f,$ and $\ddot{x}_f$ are the only components needed to generate the swingfoot trajectory as T is determined by the footstep planner. These components are determined in the below sections.

4.2. RotoGait's Stride

Because RotoGait uses both ends of each leg as feet, SLIDER effectively has four instead of two feet. As a result, the stride when using RotoGait (*Figure 6*) differs from that of normal gait (*Figure 7*).

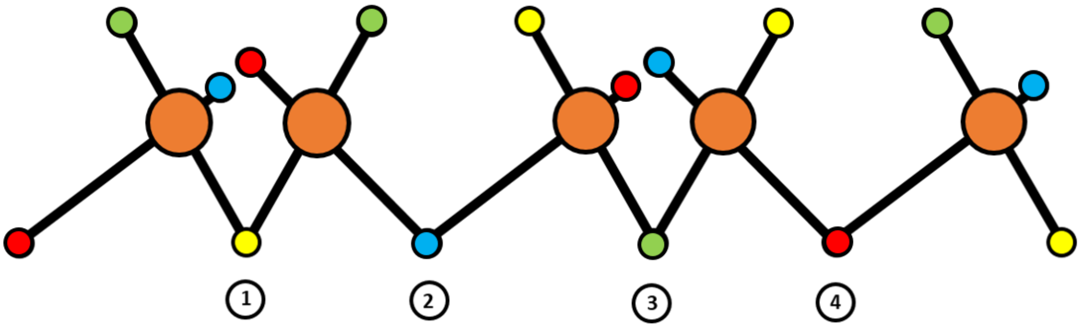

Figure 6 - RotoGait's stride

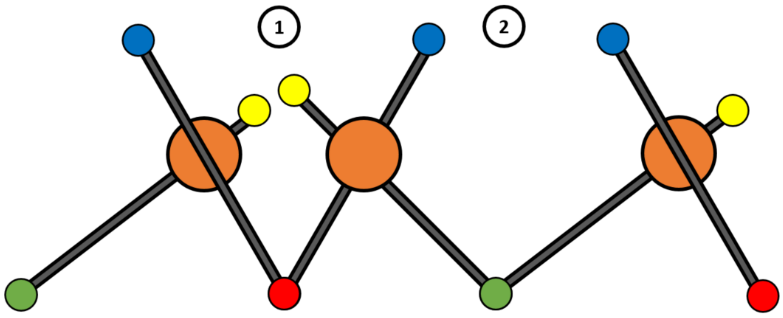

Figure 7 - Normal gait's stride

The same feet are being controlled (red and green) in both gaits, leading to two different trajectories for the swingfoot. The trajectory configuration one in *Figure 4* is when the controlled foot (red) goes up in order to get the other end of the leg (blue) in contact with the ground at the planned footstep position. In this configuration the initial movement conditions are known and the final movement conditions must be found. Once the blue foot is on the ground, that leg becomes the support leg, and the other leg becomes the swing leg. The second trajectory configuration in *Figure 8* can then be considered, where the controlled foot (green) is brought down to the planned footstep position. In this case the final movement conditions are known and the initial movement conditions are to be determined. These conditions are determined in the next sections. A flag is used to keep track of which foot is in contact with the ground to know what configuration to use for the current swingfoot movement.

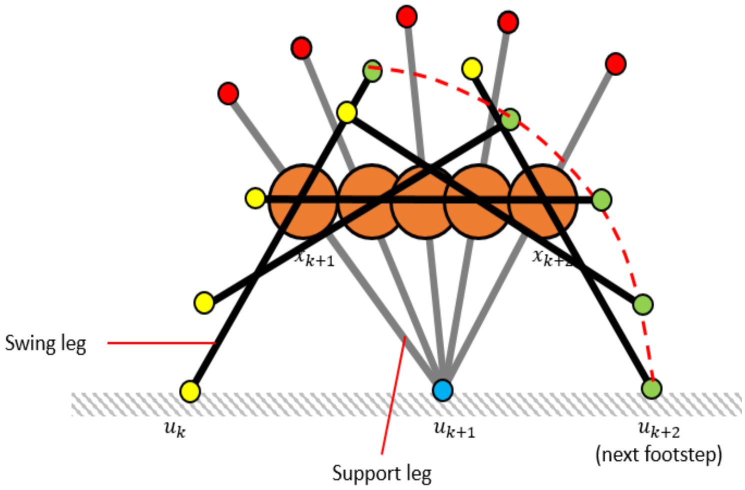

Figure 8 - Diagram of RotoGait's trajectory (Configuration two)

4.3. Configuration one: determining the final movement conditions

For this configuration, the initial conditions are known and are the same as for normal gait. To compute the trajectory, the final conditions $(x_{T\_roto}, y_{T\_roto}, z_{T\_roto})$, v_{T_roto} and a_{T_roto} as described in *Figure 9* are needed.

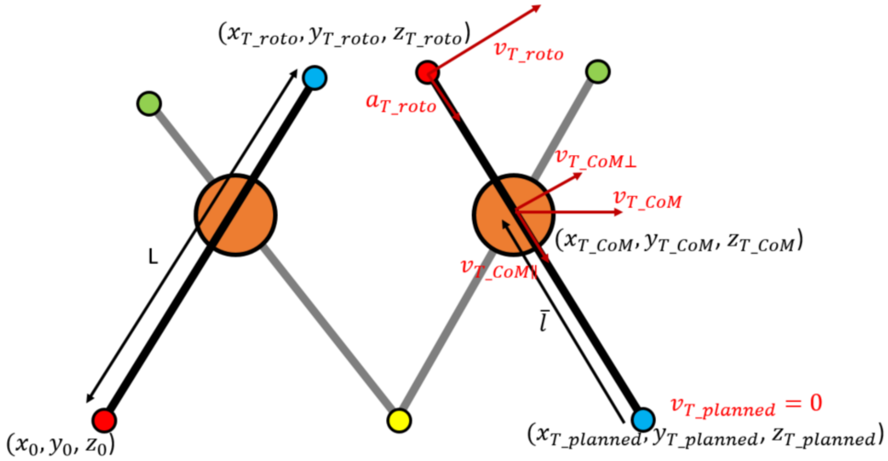

Figure 9 - Schematic of the final conditions of the movement

As the position and velocity of the blue foot are determined by the footstep planner, and the CoM position and velocity can be determined by the LIP model (see section 4.5 below), the needed final position, velocity and acceleration of the red foot are computed as following:

$$\begin{bmatrix} x_{T\_roto} \\ y_{T\_roto} \\ z_{T\_roto} \end{bmatrix} = \begin{bmatrix} x_{T\_planned} \\ y_{T\_planned} \\ z_{T\_planned} \end{bmatrix} + L * \frac{\bar{l}}{|\bar{l}|} \tag{10}$$

$$v_{T\_roto} = \frac{L}{|\bar{l}|} * v_{T\_CoM\perp} \tag{11}$$

$$a_{T\_roto} = \frac{|v_{T\_roto}|^2}{L} * \frac{-\bar{l}}{|\bar{l}|} \tag{12}$$

With $\bar{l} = \begin{bmatrix} x_{T\_CoM} - x_{T\_planned} \\ y_{T\_CoM} - y_{T\_planned} \\ z_{T\_CoM} - z_{T\_planned} \end{bmatrix}$, $v_{T\_CoM\perp} = v_{T\_CoM} - v_{T\_CoM\parallel}$ and $v_{T\_CoM\parallel} = -\bar{l} * |v_{T\_CoM}| * \frac{-\bar{l} * v_{T\_CoM}}{|\bar{l}| * |v_{T\_CoM}|}$ (13)

These final conditions are used to generate the trajectory followed by the swingfoot as shown in *Figure 4*.

4.4. Configuration two: determining the initial movement conditions

For this configuration, the final conditions are known and are planned by the footstep planner. To compute the trajectory, the initial conditions $(x_{0\_roto}, y_{0\_roto}, z_{0\_roto})$, v_{0_roto} and a_{0_roto} following the notations in *Figure 10* are needed.

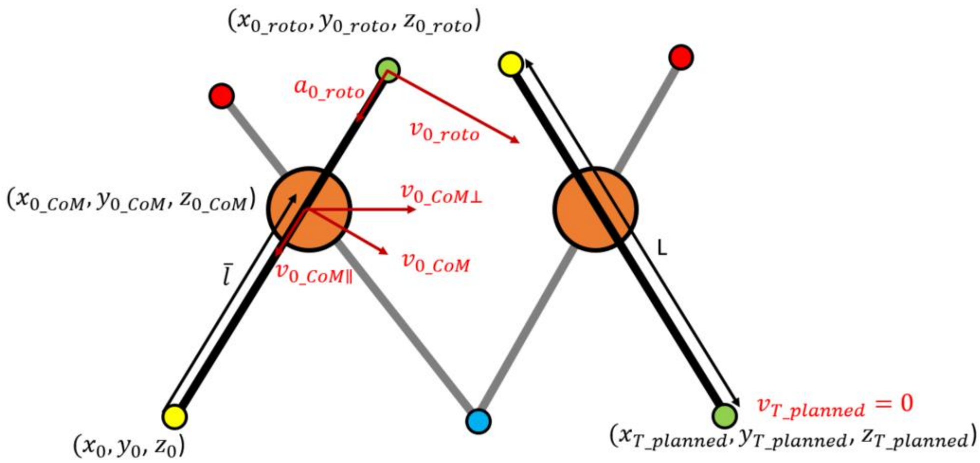

Figure 10 - Schematic of the initial conditions of the movement

The needed initial position, velocity and acceleration of the green foot are determined similarly as in configuration one knowing the position of the foot in contact with the ground (yellow) and determining the CoM position and velocity with the LIP model (see section 4.5 below):

$$\begin{bmatrix} x_{0\_roto} \\ y_{0\_roto} \\ z_{0\_roto} \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} + L * \frac{\bar{l}}{|\bar{l}|} \tag{14}$$

$$v_{0\_roto} = \frac{L}{|\bar{l}|} * v_{0\_CoM\perp} \tag{15}$$

$$a_{0\_roto} = \frac{|v_{0\_roto}|^2}{L} * \frac{-\bar{l}}{|\bar{l}|} \tag{16}$$

With $\bar{l} = \begin{bmatrix} x_{0\_CoM} - x_0 \\ y_{0\_CoM} - y_0 \\ z_{0\_CoM} - z_0 \end{bmatrix}$, $v_{0\_CoM\perp} = v_{0\_CoM} - v_{0_{CoM\parallel}}$, $v_{0\_CoM\parallel} = -\bar{l} * |v_{0\_CoM}| * \frac{-\bar{l} * v_{0\_CoM}}{|\bar{l}| * |v_{0\_CoM}|}$ (17)

4.5. LIP model

Considering that the robot has CoM position (x, y), fixed CoM height h and foot position (u_x, u_y), the LIP model is defined as:

$$\ddot{x} = \frac{g}{h}(x - u_x) \text{ and } \ddot{y} = \frac{g}{h}(y - u_y) \tag{18}$$

From this, using Euler's method we can also determine the next velocity and position of the CoM as $x = x_{previous} + \dot{x} * dt$ (19) and $\dot{x} = \dot{x}_{previous} + \ddot{x} * dt$ (20) and identical equations for y. This LIP model is used to determine the new end position of the foot when using RotoGait (see section 4.3 above).

4.6. Mid-point addition

Because a fifth order polynomial is used to determine the trajectory between the start and end positions of the foot, the computed trajectory shown in *Figure 11* would not be advantageous for SLIDER. This trajectory would indeed force the robot to use a lot the slide joint at first as X and Y don't vary much until the end and Z varies a lot, and to use the pitch only at the end of the motion. Because of this it is better to add a X mid-point to force the trajectory to be more "rounded". The Y and Z positions are still directly determined for the whole step, but the X trajectory is twofold. For configuration one, from 0 to half the step time, X goes from its initial position to initial position-0.4m and for the second half of the step goes from initial position-0.4 to RotoGait end position. Similarly for configuration two, X goes from RotoGait initial position to RotoGait initial position+0.4m then to final position. This allows to obtain the trajectory as shown in the section 5.1, which uses the pitch joint just as much as the trajectory without mid-point but uses less the slide joint, hence being more advantageous for SLIDER.

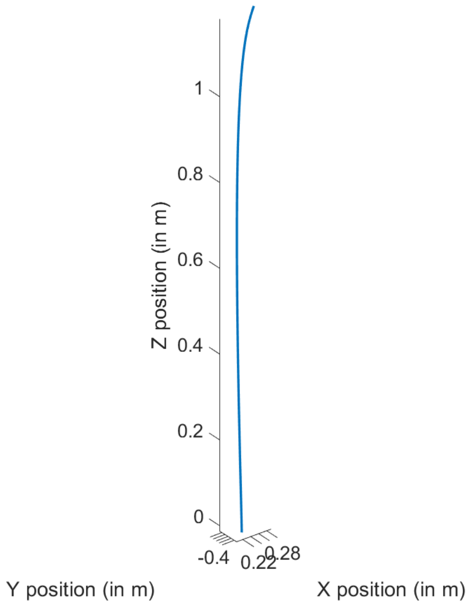

Figure 11 - Trajectory computed without a mid-point

4.7. Footstep planner modification

The footstep planner currently used optimizes the next foot position and timing based on the current positions of the CoM and both feet and the behaviour of SLIDER for normal walking. Because of that, if the actual positions of the feet are used in the footstep planner while SLIDER uses RotoGait, it will fail to optimize properly. As it is, RotoGait has been implemented to use the position and timing planned by a normal walking scenario and from there compute the new position needed by RotoGait. Therefore if the footstep planner starts optimizing differently because it takes into account the actual positions of the feet while RotoGaiting, then SLIDER will fail.

To counteract this, a separate "ghost" swingfoot trajectory is computed. This ghost trajectory follows the normal gait and feeds back normal gait swingfoot positions instead of the actual swingfoot positions to the footstep planner but is not used by SLIDER to actually move. Using this ghost swingfoot trajectory allows SLIDER to follow RotoGait while having the footstep planner still predict foot position and timing of normal gait.

4.8. Position controller

A separate position controller has been made to demonstrate the working of RotoGait. In this position controller, the CoM has been constrained to be only able to move in X direction, the Roll and Slide joints' torques of each leg are equal to zero. Using the *Figure 12*, the Pitch of the support foot and the swingfoot are defined as following:

$$q_{pitch\_support} = \frac{\theta * current\_time}{step\_time} \text{ and } q_{pitch\_swing} = \frac{(\pi-\theta) * current\_time}{step\_time} \quad (21)$$

Where step_time is the time to take a full step as determined by the footstep planner, current_time is the time already taken in that step and $\theta = 2\sin^{-1}\frac{du}{2*L}$ (22) with du the difference between the current position of the foot and the next position predicted by the planner.

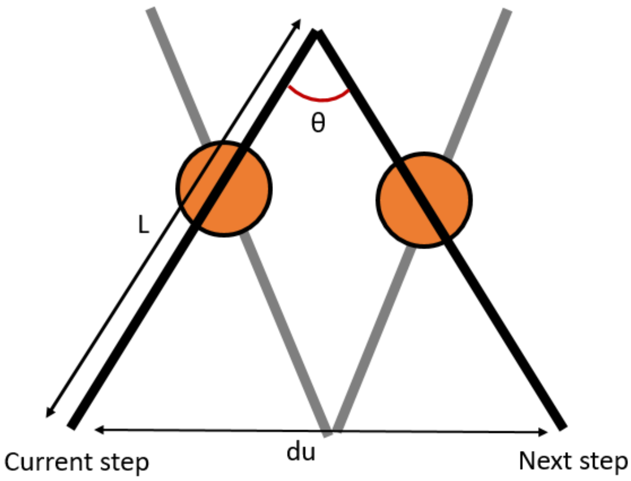

Figure 12 - Diagram of torque command elements

4.9. Power consumption

To compare the efficiency of using RotoGait to that of normal gait and determine if RotoGait can allow to use less power at certain velocities, and determine those velocities, the power consumption of both gaits has been computed for different scenarios. The energy at each time interval of the movement has been determined with the equation (23), then the total energy over the whole movement has been divided by the time taken to complete the movement to get the power consumption of the gait.

$$E = \frac{1}{2} * \left(v_{CoM}{}^{T} \cdot m \cdot v_{CoM} + w_{legs}{}^{T} \cdot I \cdot w_{legs} + v_{slide}{}^{T} \cdot m_{legs} \cdot v_{slide}\right) \quad (23)$$

The CoM velocity is determined by the LIP model (as described in section 4.5). v_{slide} is the sum of the sliding velocity of both legs, the sliding velocity of one leg being determined as the difference between the foot to CoM vector at two time intervals divided by the time interval. The position of the CoM and the foot are determined by the LIP model and the trajectory of the foot respectively. w_{legs} is the sum the rotational speed of both legs which are each computed following equation (24)from v_{leg} the velocity of the leg obtained from the trajectory generation, with r the foot to CoM vector.

$$w_{legs} = \frac{1}{r^2} * v_{leg} \cdot r \quad (24)$$

The power consumption of normal walking has been determined for 10 velocities from 0.1m/s to 1m/s. The power consumption of RotoGait has been determined for the same speeds for 6 different scenarios. The first scenario is with the same step size as normal gait (0.06m) and the mid-point of RotoGait trajectory as explained above in section 4.6. As RotoGait allows for bigger step sizes, two other step sizes have been tested, 0.15m and 0.3m. Because using a mid-point of initial position-0.4m is not necessarily useful for bigger footsteps as the trajectory generated is closer to the one illustrated in *Figure 4* and *Figure 8*, two other scenarios for step size of 0.3m are with a mid-point of initial position-0.2m and initial position-0.1m along X. Finally, RotoGait can work with the CoM height being at 0.6m instead of 0.7m, which means that the CoM is half-way from both ends of the legs, allowing to remove the slide motion of the legs and therefore consume less power. This last scenario has therefore also been tested for the different velocities.

5. Results

5.1. RotoGait trajectory

The *Figure 13* and *Figure 14* show the trajectory followed by the swingfoot during RotoGait compared to normal gait for a full stride of RotoGait(2 steps for each leg).

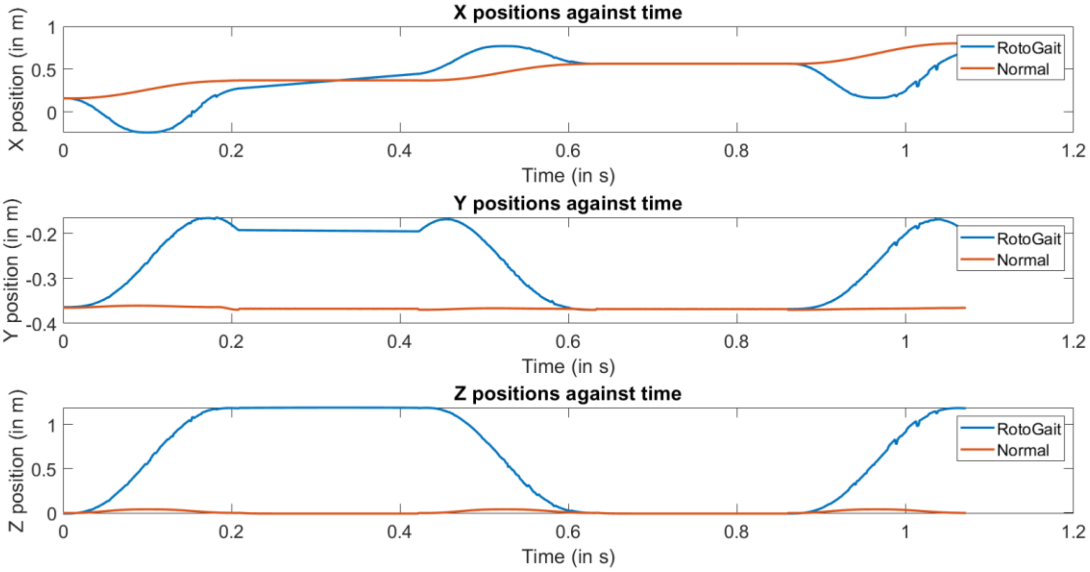

Figure 13 - X,Y and Z positions against time for both RotoGait and normal gait

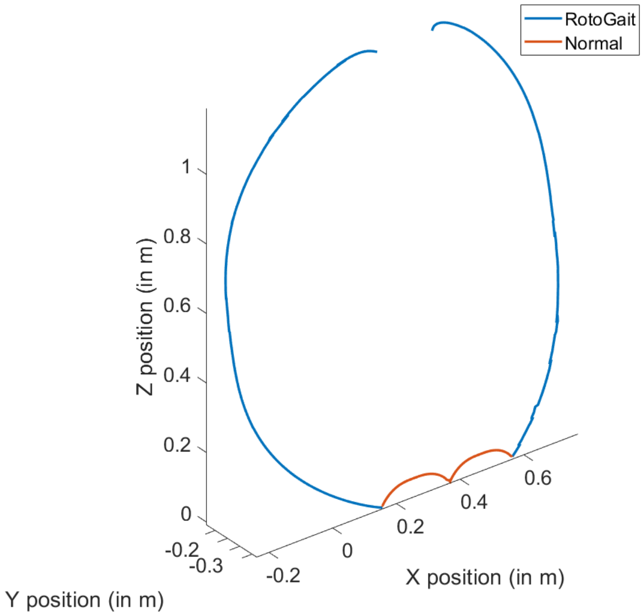

Figure 14 - X,Y,Z trajectory of the swingfoot for RotoGait and normal gait

5.2. Whole body control

The major issue encountered using the Whole body controller to run RotoGait is represented in *Figure 15*, showing that the leg starts to rotate and then changes rotation direction when reaching the horizontal. For the full video of the movement please see Whole_body_controller.mov at the following link:
https://drive.google.com/drive/folders/1WloMcn5UL1ysusvFWfCNXPUTaMC0B9C1?usp=sharing

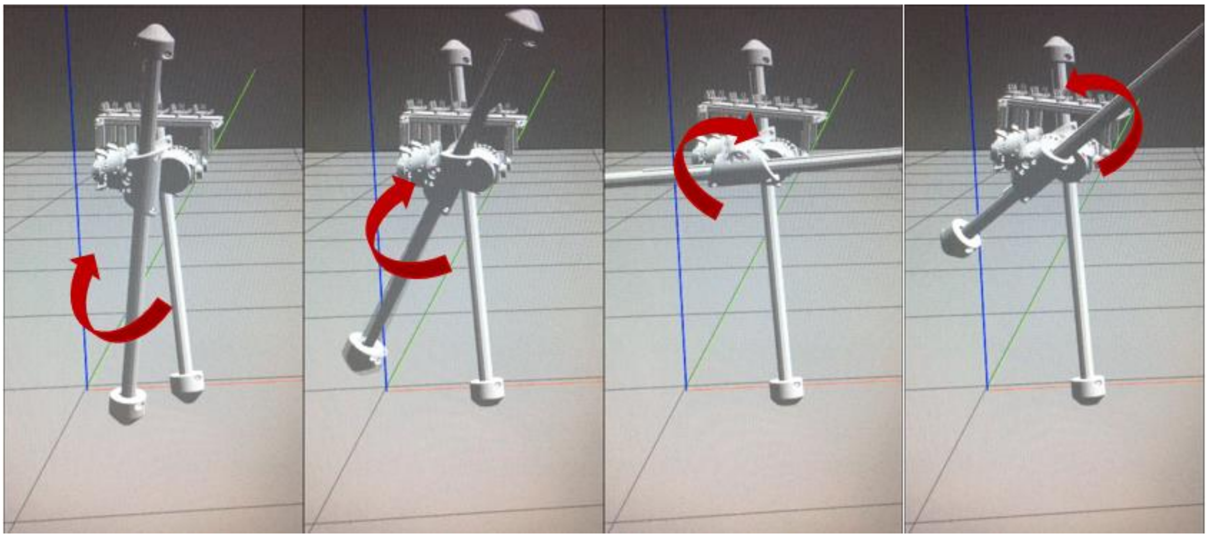

Figure 15 - RotoGait using the whole body controller

5.3. Position control

The simulation of SLIDER using RotoGait by position control is illustrated in *Figure 16*, for the full video of the movement please see Position_control.mov at the following link: https://drive.google.com/drive/folders/1WloMcn5UL1ysusvFWfCNXPUTaMC0B9C1?usp=sharing

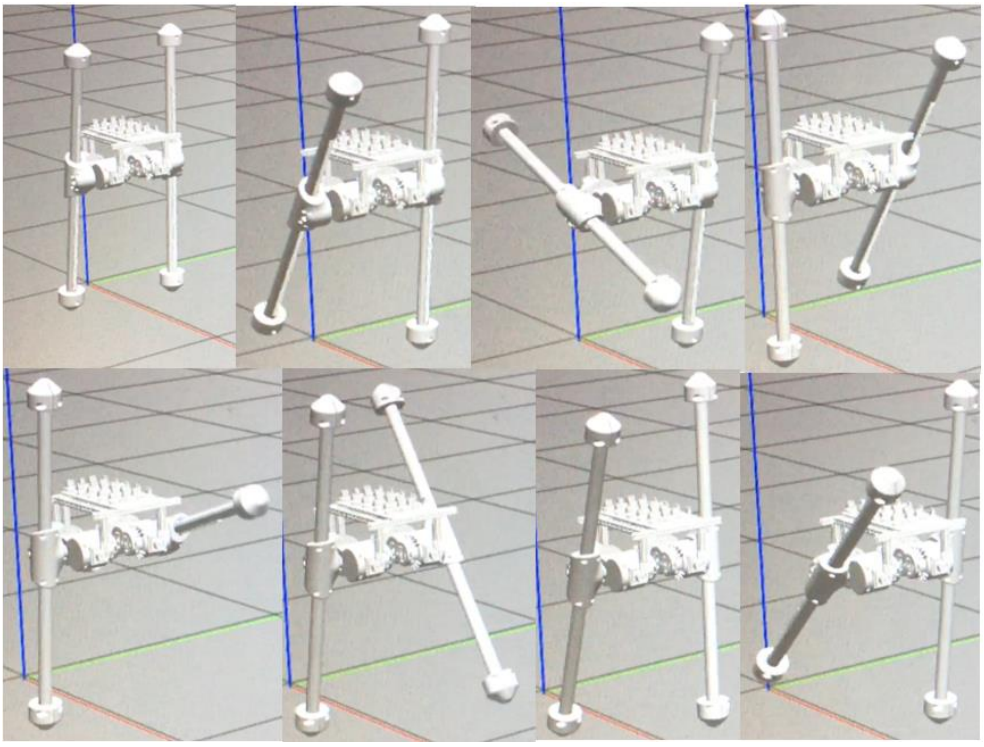

Figure 16 - RotoGait by position control

5.4. Power consumption

The power consumption for 10 different walking speeds are displayed in *Figure 17* for normal gait, RotoGait and RotoGait with a CoM height at 0.6m to remove the sliding of the leg and therefore use less power. RotoGait consumes more power than normal gait for every tested velocity, with the gap between power consumptions increasing enormously at higher velocities. Even though having the CoM at a 0.6m height does allow SLIDER to consume less power than for the CoM at 0.7m when using RotoGait, this difference is not remarkable.

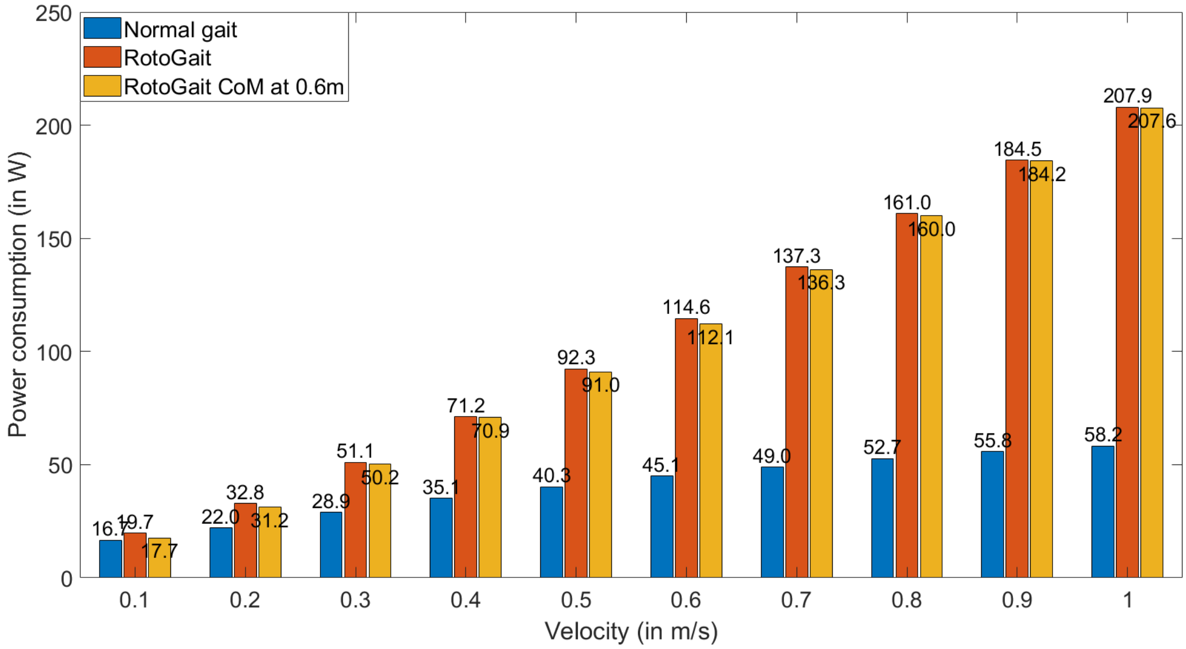

Figure 17 - Comparison of power consumption for normal gait, RotoGait and RotoGait without slide power consumption

Figure 18 shows the difference of power consumption for RotoGait for the same velocities as before for different step sizes. For a step size of 0.3m, SLIDER fails at 0.1m/s and 0.2m/s. At lower velocities, a bigger step size consumes a lot more power than a smaller steps, but the power consumption of bigger step sizes decreases quickly at higher velocities and becomes more power efficient than smaller step sizes at high speeds.

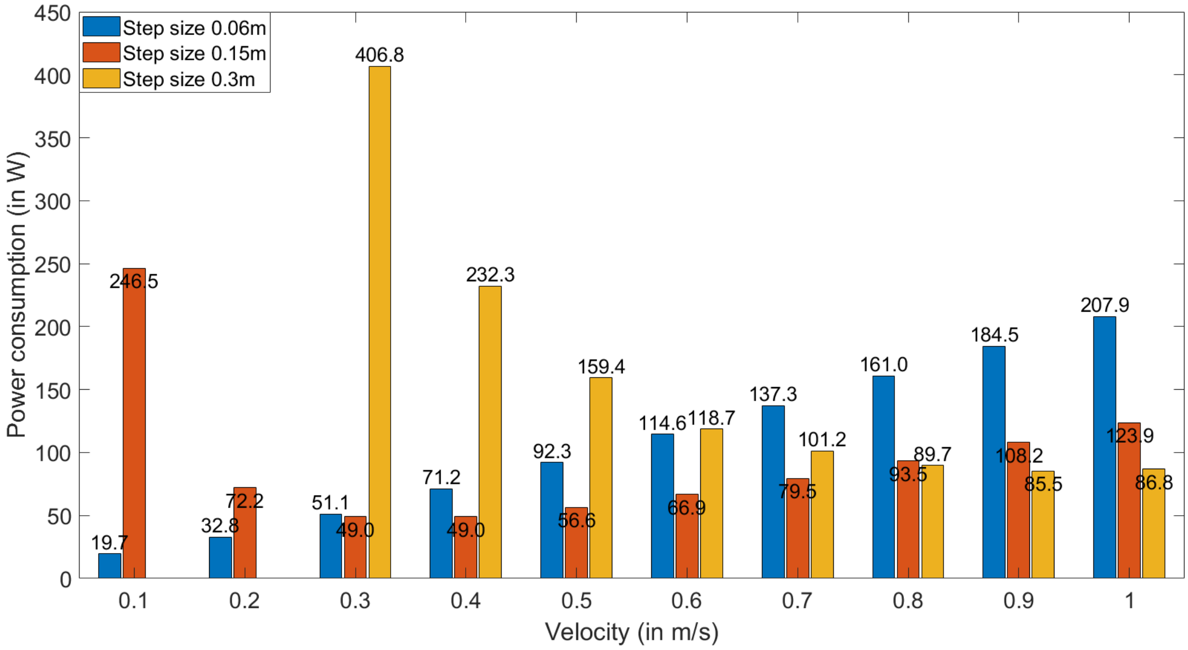

Figure 18 - RotoGait power consumption for 3 step sizes: 0.06m (like normal gait), 0.15m and 0.3m

For a bigger step size (here 0.3m), reducing the distance between the initial X position and the mid-point X position allows to consume less power as shown in *Figure 19*.

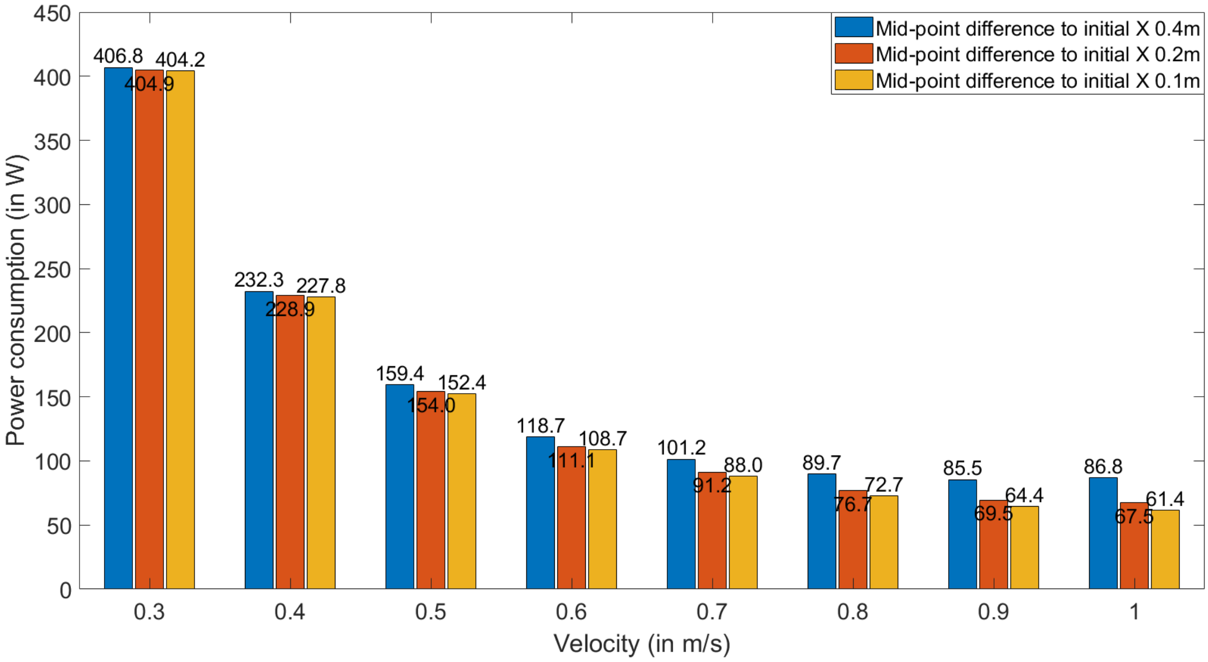

Figure 19 - RotoGait power consumption for step size of 0.3m for different trajectory mid-point positions

6. Discussion

6.1. RotoGait trajectories

As expected, the major difference between RotoGait and normal gait is the Z position as shown by *Figure 13* and *Figure 14* as the controlled foot goes all the way over the robot allowing RotoGait to happen. The X position in RotoGait follows the trend of the one from normal gait showing an overall forward movement of the leg. Nevertheless, because of the addition of a mid-point on the swingfoot trajectory, the X position of RotoGait will vary more than normal walking before reaching the end X position. Furthermore, the X trajectory when the leg is support leg will be different from configuration one to two as from two to one. Indeed as the controlled foot is at the top between configurations one and two it will rotate around the other end of the leg therefore linearly increasing X position. Between configurations two and one on the other hand, the controlled foot is in contact with the ground and X position therefore doesn't change. Because of the V shape taken by the legs of SLIDER during normal walking, there is also a variation in Y position during RotoGait compared to normal gait as the foot moves from the lower and inner part of the V shape to the top and outer part of the V shape, and vice-versa. This variation in Y position could be reduced as the V-shape of the leg may not be necessary when SLIDER RotoGaits.

6.2. Footstep planner modifications

Using a "ghost" trajectory to feed back to the footstep planner normal walking positions instead of the actual positions, which differ from normal for RotoGait, does allow to continuously predict normal gait optimized position and timing. As a result, the prediction are separated from the robot movement consequently allowing SLIDER to use RotoGait without interfering with the footstep planner which would lead to the failure of the robot as the optimization of the footstep planner is based on normal walking.

Nevertheless a consequence of this modification is that the planning of SLIDER's behaviour is separated from its actual behaviour. The robot will thus not be able to compensate for errors. If the trajectory actually implemented doesn't follow the desired trajectory then, as there is no feedback to the planner, the robot will not be able to correct for that mistake. Similarly if there is an unpredictable movement, such as a push, or a slope, there again since there is no feedback of what is actually happening, the planner will still consider normal flat ground walking without knowing there is a change, resulting in SLIDER to fail to recover from that environment change.

6.3. Whole body control

The Whole body controller is the major hinderer of the successful implementation of RotoGait following SLIDER's control diagram from *Figure 2*. As shown in *Figure 15* and in Whole_body_controller.mov, while using the Whole body controller to determine the joints' torques, the swing leg fails to fully rotate even though the desired trajectory is correctly computed as in section 5.1. The leg rotates correctly to approximately the horizontal plane but then the rotation changes direction, stopping the full rotation of the leg and therefore leading to the failure of SLIDER as it is not able to follow the desired trajectory. This is a consequence of the change of sign in the torque of the pitch when the leg crosses the horizontal plane instructing the leg to change the direction of rotation. This change of sign of the pitch torque is the result of the functioning of the Whole body controller.

The controller is indeed based on the inverse dynamics of normal gait and also fails to consider differences in the two gaits as the four possible feet in RotoGait which should be considered in the force selection matrix of the controller for example. As a consequence, since the whole body controller is trying to compute torque instructions based on the wrong dynamics for RotoGait the torque predicted is consistent for a normal gait scenario and not for RotoGait, leading to the failure of RotoGait.

As determining the dynamics of RotoGait was part of the project of another student, Zac Galibov, the Whole body controller can be modified in the future to be based on the inverse dynamics specific to RotoGait and not normal gait. This should thus allow the right torque instructions to be computed by the Whole body controller and allow SLIDER to RotoGait.

6.4. Position control

It is because of the shortcomings of the Whole body controller that the position controller has been developed to show that RotoGait is indeed a movement that SLIDER can produce given the right torque instructions. The results from the position control shown in *Figure 16* and more particularly in Position_control.mov allows to demonstrate the principle of RotoGait and proves it is applicable to SLIDER. Nevertheless, this is a simplified implementation of RotoGait that cannot be used in real-life situations. This is because of the different constraints applied in this control and the fact that it does not take into account the dynamics of the robot. It would therefore not be a stable gait in other contexts or outside of simulation on the physical robot. Furthermore, because of the constraints and as it doesn't work like normal gait, any further results stemming from this controller would not be comparable to results coming from the Whole body controller on normal gait. The sole purpose of this position control is hence to demonstrate the movement of SLIDER when it uses RotoGait.

6.5. Power consumption

The *Figure 17* shows that the power consumption of RotoGait increases too much at higher velocities to be more efficient than normal gait, at least for the tested velocities. This is because for a fixed step size (the same as normal gait at 0.06m), the higher the speed is, the higher the angular velocity of the leg needs to be to take the exact same step. If the angular velocity is higher, then the power consumption is too. Furthermore, the trajectory followed by the swingfoot already presents a low amount of sliding of the legs. Therefore, decreasing the CoM height to 0.6m, thus removing completely the need of sliding, is actually just slightly more efficient than with the CoM at 0.7m.

Nevertheless, one major advantage of RotoGait is to be able to have a bigger step size than for normal walking. This is why different step sizes where tested for the same 10 velocities. Firstly, for low speeds, a bigger step leads to the failure of SLIDER which is why no power consumption has been reported for a step size of 0.3m at 0.1m/s and 0.2m/s in *Figure 18*. This can be caused by the fact that for a specific velocity, if the step size is too big, the ZMP might not be able to stay in its support polygon as the CoM starting position is too far behind the support foot and its end position too far past the support foot. For velocities where SLIDER does not fail, the lower velocities consume more power than higher velocities contrary to smaller step sizes where low velocity consume less power than high velocities. At lower speed, the support leg's pitch still uses up a lot of power to balance the CoM from one side to the other while at higher speeds SLIDER can reach a dynamic stability allowing to use less power to walk. In addition, for bigger step size, at a same velocity, the angle to cover by the swing leg is smaller than for small step sizes and the time to cover that angle is bigger. As a result the angular velocity of the pitch joint is smaller at bigger step sizes leading to less power being consumed at higher velocities.

Finally, *Figure 19* shows that for the same step size and velocity, using another trajectory can improve or worsen the power consumption of SLIDER. In this case, as the step is big, decreasing the difference of X position between the mid-point and the initial position of the swingfoot allows to use less power.

Nevertheless, none of the tested scenarios of RotoGait was more power efficient than normal gait at the same velocity. The fact that the power consumption of RotoGait can be decreased from 207.9W to 61.4W for 1m/s, getting it close to the 58.2W power consumption of normal gait shows however that with an optimal step size and trajectory, RotoGait could be more efficient than normal gait at higher velocities.

7. Conclusion

As a conclusion, this project has shown that while SLIDER can physically use RotoGait instead of its normal gait, SLIDER's current controller is not adapted to allow the use of RotoGait. Because of this, the fastest velocity reachable with RotoGait has not been tested. Furthermore, the estimation of the power consumption of both normal gait and RotoGait shows that RotoGait is not more efficient than normal gait for the tested velocities and trajectories. Despite this, the results give good grounds for expecting that RotoGait would be more efficient than normal gait if the trajectory is optimized.

In the future, this work should be combined to the work of Zac Galibov to modify the Whole body controller to consider RotoGait's dynamics therefore allowing to implement RotoGait and test the maximum velocity reachable. The trajectory should also be optimized to allow for a better power efficiency of RotoGait. Finally, the footstep planner could also be modified to optimize footstep positions based on RotoGait positions as well so that it can take bigger steps and use feedback from SLIDER's actual positions instead of using the "ghost" trajectory.

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